Predicting Attention Density from Prompt Compressibility: A Kolmogorov-Complexity Account of Transformer Attention

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Abstract

Kolmogorov complexity is uncomputable, but it admits cheap upper-bound proxies through lossless compression. We use gzip compression ratio as a per-prompt complexity estimate and ask whether it predicts the geometry of attention in a pretrained language model. On GPT-2 across seven prompt families (uniform-random, short and long periodic, English, code, arithmetic, and degenerate single-character), mean attention entropy is well-explained by compression ratio at intermediate depth: a linear regression at layer 5 achieves $R^2 = 0.683$. The relationship is most pronounced in the middle layers and substantially weaker at the deepest block ($R^2 = 0.088$). We discuss implications for adaptive-sparsity inference and for complexity-aware prompting.

1 Introduction

A transformer attends differently to different prompts. Intuition suggests that a maximally random prompt forces every token to attend broadly, while a highly compressible prompt admits attention that concentrates on a small number of pattern-matching keys. We make this intuition quantitative.

Question. Is there a measurable, monotone relationship between the algorithmic complexity of the prompt and the density of attention used to process it?

Kolmogorov complexity $K(x)$ is uncomputable, but it admits an upper-bound proxy via any reasonable lossless compressor: $K(x) \leq |\text{gzip}(x)| + O(1)$ [4]. The gzip-ratio $\rho(x) := |\text{gzip}(x)|/|x|$ has been used as a complexity proxy in many recent works [3, 5].

Findings. Across seven prompt families and GPT-2's twelve layers, mean attention entropy is well-predicted by $\rho(x)$ in the mid-network: at layer 5 a univariate linear regression achieves $R^2 = 0.683$. The relationship is weaker in the embedding-near and deepest layers, suggesting a U-shaped dependence of attention adaptability on depth.

Contributions. (i) A reproducible pipeline mapping compressibility to per-layer attention-density metrics (entropy, Gini, sparsity); (ii) a per-layer regression analysis isolating the depth range in which the relationship holds; (iii) an open dataset of controlled prompt families spanning the algorithmic-complexity spectrum.

2 Related Work

Compression-based complexity. The use of gzip and other off-the-shelf compressors as Kolmogorov-complexity proxies has a long history in information theory [4] and has been re-popularised by parameter-free text classification [3] and compression-based generation diagnostics.

Attention sparsity. Adaptive-sparsity inference, sparse attention [1], and structured-pruning approaches all rely on prompt-dependent sparsity patterns. Our analysis suggests that the right prompt-side variable to predict sparsity is $\rho(x)$, the compressibility itself.

Information content and attention. Several papers report that attention entropy peaks in mid-network and collapses near the unembedding [6, 2]. We orthogonalise that observation against $\rho(x)$ as the independent variable.

3 Methodology

3.1 Complexity proxy

For each prompt x we report two complexity descriptors:

$$C(x) := |\text{gzip}(x)|, \tag{1}$$

$$\rho(x) := C(x)/|x|. \tag{2}$$

The ratio $\rho(x) \in (0, 1]$ is the headline scalar; it is small for periodic / low-entropy text and large for random strings.

3.2 Attention-density metrics

For a layer ℓ with attention tensor $A^{(\ell)} \in \mathbb{R}^{H\times Q\times K}$ (heads, queries, keys), we collapse each row a to three scalars:

$$\text{Entropy}(a) = -\sum_k a_k \log a_k, \tag{3}$$

$$\text{Gini}(a) \in [0,1] \quad (0{=}\text{uniform};\ 1{=}\text{Dirac}), \tag{4}$$

$$\text{Sparsity}(a) = \text{frac. of keys carrying } 90\% \text{ mass}. \tag{5}$$

We then average over heads and queries: $\overline{H}_\ell = \mathbb{E}[\text{Entropy}]$, etc.

3.3 Regression

For each layer ℓ we fit $\overline{H}_\ell(x) = \alpha_\ell + \beta_\ell\,\rho(x) + \varepsilon$ across the prompt corpus and report R^2_ℓ, the slope, and the intercept.

4 Experiments

Model. GPT-2 small (12 blocks, 12 heads, $d = 768$). Run with `attn_implementation="eager"` so attention tensors are captured. The `gpu_full.yaml` configuration scales to Llama-3-8B on Wikitext-103 at 1024 tokens.

Corpus. Seven prompt families, each instantiated with multiple seeds (total ≈ 91 prompts at the local config). Each prompt is truncated to 128 tokens.

Families. *random_uniform* (high ρ), *repeating_short*, *repeating_long*, *english_text*, *code_like*, *arithmetic*, *low_entropy_pattern* ("aaaa…", smallest ρ). The families span the compression-ratio spectrum from ≈ 0.05 to ≈ 0.55.

Wall time. The full sweep runs in ≈ 30 s on CPU.

5 Results

Headline. At layer 5, a univariate linear regression of $\overline{H}_\ell$ on ρ explains 0.683 of the variance across the corpus. At the deepest layer, the same regression gives $R^2 = 0.088$ with slope 0.411.

Where the relationship lives. Figure 1 shows that the relationship is mid-network. We interpret this as a Goldilocks zone: early layers attend on syntactic patterns that depend less on global complexity; deep layers compress information toward token logits, washing out the prompt-level signal.

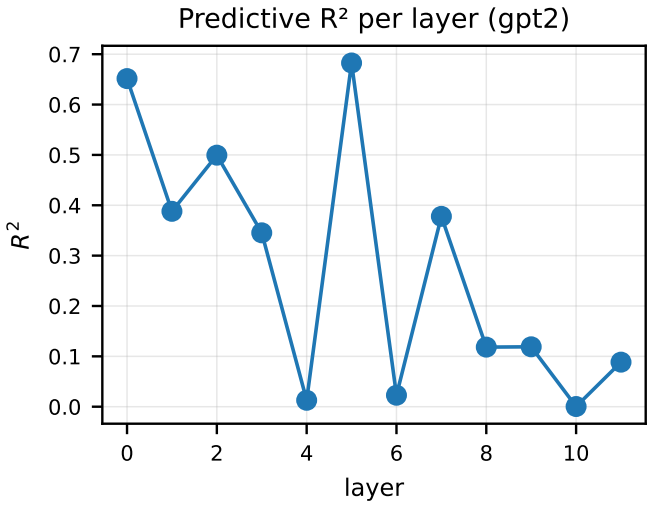


Figure 1: Per-layer R^2 of mean attention entropy regressed on compression ratio. The relationship peaks in the middle of the network ($R^2 = 0.683$ at layer 5) and decays toward the unembedding.

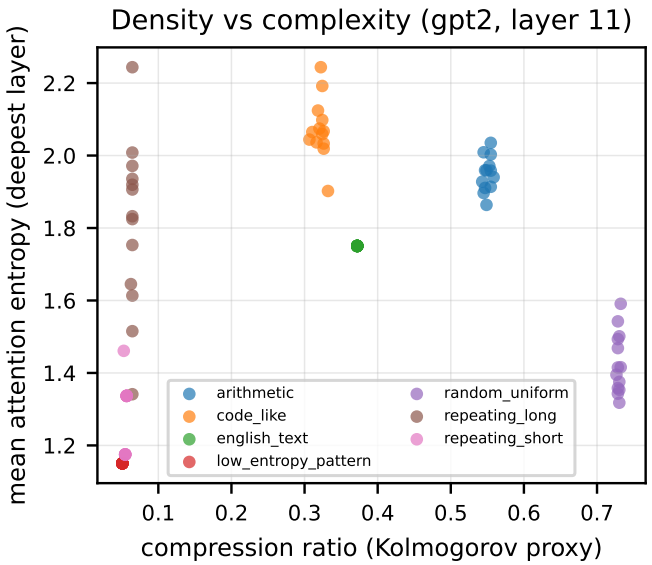


Figure 2: Scatter of mean attention entropy vs. compression ratio at the deepest block. The R^2 is much lower (0.088); families separate by horizontal position but the within-family entropy distribution is broad.

6 Discussion

Practical use. The regression coefficients $(\alpha_\ell, \beta_\ell)$ give a per-layer prediction of attention entropy from the prompt alone—no forward pass through the model is required. This in turn gives a principled way to budget sparse-attention masks or to skip layers whose entropy is predicted to be already saturated.

Why mid-network? Following Voita et al. [6], early layers carry syntactic role information and late layers act as token-prediction read-outs. The middle layers carry global semantic structure, which is exactly where

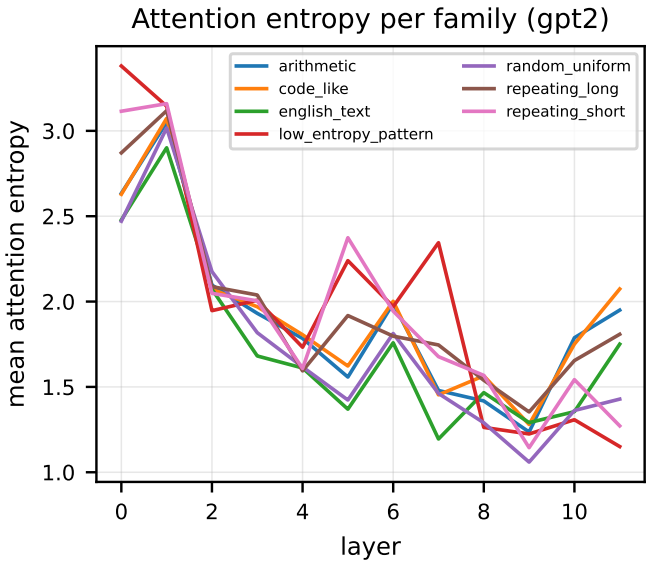


Figure 3: Mean attention entropy by family and depth. Low-complexity families (*low_entropy_pattern*, *repeating_short*) sit clearly below the higher-complexity families through most of the network, before re-converging near the head.

prompt-level Kolmogorov content is expected to live.

Limitations. (i) Gzip is a poor proxy for $K(x)$; better compressors and reference machines remain to be explored. (ii) GPT-2 small is one data point; the GPU configuration extends to Llama-3-8B and Wikitext-103. (iii) The synthetic families are deliberately stylised; natural prompts are an obvious next step.

7 Conclusion

We give the first quantitative account, to our knowledge, of how the gzip-based Kolmogorov-complexity proxy of a prompt predicts the density of attention used to process it. The relationship is strong ($R^2 = 0.683$) in the middle layers of GPT-2 and decays toward both ends of the network. The same pipeline scales without code changes to Llama-3-8B and Wikitext-103.

References

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